Agentic Economics: The Business Case

The ROI Framework for Agentic AI Adoption

What it takes to turn the $2/hour opportunity into P&L reality

A Three-Part Executive Briefing Series

The Labor Multiplier

What agents can actually do — the 67% of delivery labor outside the code editor that AI is built to handle at a fraction of human cost

Agentic Economics

How to calculate and capture ROI — agent compute costs, infrastructure investment requirements, and the honest payback math

Agentic Delivery

The operating model for safe scale — instrument infrastructure, pilot oversight, and the governance that enables the throughput multiplier

Each briefing is complete on its own. Together they form a full strategic framework for agentic AI investment.

Three Briefings, Three Decisions

The Labor Multiplier	Which of the 67% of delivery labor is ready to automate — and what organizational investment does capturing it require?	`Opportunity Decision`
Agentic Economics	What is the realistic ROI timeline, and what infrastructure makes the $2–5/hour agent arbitrage accessible at scale?	`Capital Decision`
Agentic Delivery	What governance infrastructure — instruments, oversight model, no-fly zones — must exist before agent programs can operate safely?	`Operating Model Decision`

These are the three decisions the briefing series is designed to drive. This talk addresses one of them.

The Complete Decision Framework

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Identify the Opportunity

The Labor Multiplier: 67% of delivery labor has no AI coverage — discovery, compliance, documentation, and validation are ready for agents

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Quantify the Investment

Agentic Economics: agent compute at $2–5/hour creates a structural cost arbitrage — the ROI depends on the verification infrastructure

🏗️

Enable Safe Operation

Agentic Delivery: the instrument panel, pilot model, and governance structure that make the multiplier achievable at enterprise scale

💡 The Labor Multiplier identifies what is possible. Agentic Economics quantifies the return. Agentic Delivery specifies the infrastructure that makes both achievable.

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PART 1

The Labor Arbitrage Opportunity

Agent compute at $2–5/hour vs $130/hour for senior engineers creates a structural cost differential.

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The Cost Gap

$5/hr vs $130/hr — a 20–65x fully-loaded differential

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Three Work Buckets

High-value, automatable, toil — each priced differently

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The DORA Signal

90% AI adoption, 24% output trust — the gap follows

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Agent compute cost vs senior engineer (fully loaded)
↳ $2–5/hr vs $100–130/hr
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The Cost Differential That Reframes Engineering Economics

Fully-loaded agent compute vs senior engineer labor, 2026

20–65x

fully-loaded labor cost differential — agent compute vs senior engineer

Agent: $2–5/hr (API + orchestration + monitoring) | Senior engineer: $100–130/hr (salary, benefits, overhead) — BLS 2024

🤖 Agent Compute

$2–5/hr for production-grade agentic workflows including API calls, orchestration, and monitoring

💼 Mid-Level Developer

$85–110/hr fully loaded — 17–55x more expensive than agent compute for the same task time

🌐 Offshore Developer

$25–85/hr — still 5–42x more expensive than agent compute even at the lowest offshore rates

🎯 The Strategic Question

Which engineering hours can move to the $2–5/hr tier — and what infrastructure is required?

💡 The question is not whether agents are cheaper. The question is what work can move to $2–5/hour, and what infrastructure is required to capture it.

Three Categories of Engineering Work — Three Price Points

🧠 High-Value Work

Architecture, strategy, and complex debugging — judgment-intensive work at $100–150/hr.

- Customer conversations
- Trade-off evaluation
- System design decisions

⚙️ Routine Work

Code review, test writing, documentation — pattern-based work already accessible at $2–5/hr

- Bug investigation
- Dependency updates
- Compliance checking

🔁 Pure Toil

Issue triage, audit prep, status reports — rule-following work best suited for $2–5/hr agents

- Duplicate detection
- Format enforcement
- Data aggregation

📊 DORA 2025: 90% of organizations have adopted AI tools — only 24% trust AI outputs enough to deploy without extensive review. The capture problem is structural.

PART 2

The Capture Problem

AI pilots are approved and the P&L is unchanged. Five barriers — one root cause: no observability infrastructure.

📉

McKinsey Validation

5% of pilots reach material P&L improvement

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The Perception Gap

Teams believe 20% faster; measured 19% slower

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Five Barriers, One Root

No instrumentation means no capture

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McKinsey QuantumBlack 2024 — AI pilot success rate
↳ 5% deliver material P&L improvement
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Most Organizations Have Been Cautious — Correctly

McKinsey QuantumBlack 2024: P&L impact rate for AI pilots across organizations

5%

of AI pilots deliver material improvement to organizational P&L

McKinsey QuantumBlack: The State of AI in 2024 — a half decade in review

✓ The Correct Instinct

Caution about AI ROI has been proportionate — most pilots have not delivered what they projected

🔬 A Diagnostic, Not a Verdict

The 95% shortfall identifies where the model breaks down — a blueprint for what to fix

🏗 Infrastructure Is the Variable

Pilots reaching the 5% share one common factor: automated verification before agents touch code.

💡 The gap between pilots and P&L impact is not model quality. It is measurement infrastructure — teams are flying without instruments.

Flying Blind AND Confident

METR RCT 2025: experienced developers on complex real-world open-source tasks

20% → −19%

self-reported productivity gain vs independently measured slowdown on complex tasks

METR RCT 2025 — experienced open-source developers; fewer than 44% of AI suggestions accepted on complex repos

🎭 The Perception Gap

Developers self-reported 20% faster. Measured: 19% slower on complex, unfamiliar codebases.

🔍 Where the Slowdown Occurs

Iteration overhead explains the gap. Fewer than 44% of suggestions accepted on complex repos.

✅ Where Gains Are Reliable

Well-scoped routine tasks with clear requirements and automated verification show consistent gains.

💡 Organizations without measurement infrastructure cannot distinguish confidence from performance — and cannot course-correct when pilots underdeliver.

Five Barriers With One Root Cause

The most common reasons AI investment does not reach the P&L — all trace to missing infrastructure

Verification	No automated tests means every agent output needs full human review.	`Verification Gap`
Tacit Context	Undocumented context is invisible to agents — wrong assumptions follow	`Context Gap`
Review Gates	An agent done in 30 minutes still waits 2 days in the review queue.	`Flow Gap`
No Guardrails	Every task requires negotiating scope before the agent can start.	`Governance Gap`
Fragmentation	Context scattered across tools means agents get partial inputs.	`Integration Gap`

All five barriers trace to one diagnosis: no observability infrastructure. That is a solvable problem.

Governance Infrastructure Pays for Itself

❌ Shadow AI Risk

- 20% of 2025 data breaches originated from shadow AI usage
- Unauthorized models bypass security controls and audit trails
- **$670K additional cost**
  Per incident involving shadow AI vs standard breach (IBM 2025)

💡 AI Security Controls

- Sanctioned model registry with access policy enforcement
- Automated scanning integrated in CI/CD pipelines
- Usage monitoring with anomaly detection and audit trails

✅ Measurable Protection

- Security investment reduces both breach probability and magnitude
- Governance becomes self-funding at the first prevented incident

$2.2M	$4.44M
avg savings with AI security controls	global avg breach cost (IBM 2025)

PART 3

Quick Wins: The Issue Lifecycle Pattern

One workflow, one payback number: issue lifecycle automation proves ROI before the full platform investment.

⚡
3.6-Day Payback
$1M savings, 3.6-day payback — single workflow

🔑
The Wedge Principle
Start where work is frequent and approval-light

🔄
Self-Funding ROI
Quick wins generate budget for full transformation

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Issue lifecycle automation — 50-person team scenario
↳ $1,001,000/year savings · 3.6-day payback
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A Single Automatable Workflow Pays Back in 3.6 Days

Issue lifecycle: research → planning → execution → review (50-person team scenario)

3.6 days

payback period for full issue lifecycle automation — 50-person team scenario

Internal scenario: 20 issues/week, $100/hr avg loaded rate, agent cost $3/hr. See README for full assumptions.

📥 Current State (manual)

$1,040,000/year — 20 issues/week × 10 hrs each × $100/hr loaded labor rate

🤖 With Lifecycle Agents

$39,000/year total — $7,800 agent API costs plus $31,200 human validation time

💰 Annual Savings

$1,001,000/year — 96% reduction in total workflow cost for the same output volume

🏗 Infrastructure Required

4–6 hours of one-time setup. No repository restructuring or CI/CD rewrite required.

💡 The $10,200 first-year investment generates $1,001,000 in savings — the highest-ROI entry point into agentic economics for most engineering organizations.

From Manual Lifecycle to Agent-Assisted Execution

Four stages, dramatically compressed

❌ Manual Workflow

1. **Issue triage**
   30 min @ $100/hr = $50/issue
2. **Planning**
   4 hrs research + spec = $400/issue
3. **Implementation**
   6 hrs coding @ $100/hr = $600
4. **Code review**
   60 min @ $100/hr = $100

⏳ $1,150/issue fully loaded — 11.5 hours per issue lifecycle

✅ With Lifecycle Agents

1. **Automated triage**
   5 min agent @ $3/hr = $0.25/issue
2. **Agent-drafted plan**
   30 min + 5 min approval = $3.50/issue
3. **Agent implementation**
   90 min @ $3/hr + 15 min review = $29.50
4. **Automated review**
   5 min human validation = $8.33

✓ $41.58/issue — 96% cost reduction per issue

$1,040,000/year → $39,000/year for 20 issues/week

Three Principles That Make the Wedge Work

🎯 The Wedge

Start where work is frequent, measurable, and approval-light. Issue triage meets all three criteria.

- High volume drives fast proof
- Measurable before/after baseline
- Approval-light reduces bottlenecks

💰 Self-Funding ROI

Quick wins generate the budget for larger infrastructure. $1M in savings funds the $1M buildout.

- Phase 1 funds Phase 2
- Executive buy-in from real data
- Team competency as a byproduct

🛡️ Governance as Gate

IBM data: organizations with governance before scaling see $2.2M in breach-cost savings.

- Audit trails from day one
- Access policy before expansion
- Shadow AI risk eliminated early

📖 See The Journey to Agentic SDLC for step-by-step setup instructions with working code examples for each stage of the issue lifecycle.

PART 4

The Leadership Calculation

Year 1 is negative, Year 2 is modeled positive. Infrastructure investment precedes agent deployment for reliable ROI.

📉
The Modeled J-Curve
Year 1 ~-$425K, modeled 240% 3-year ROI

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Dashboard Checkpoint
Directional cohort signal — not realized savings

🎯
Three Leadership Asks
Instrumentation, governance, workflow redesign

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Modeled 3-year ROI — 50 engineers, $1M infrastructure investment
↳ Year 1: ~-$425K → Year 2: modeled positive → Year 3: modeled 240% ROI
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The J-Curve: Year 1 Is Negative

Modeled — 50-person team, $100/hr average

Year 1	Year 2	TARGET Year 3
Investment Phase	**Returns Phase**	**Compounding**
~$1M investment. $635K savings. Net: ~-$425K.	$1.486M run rate. No additional investment. Net: modeled positive.	Net: ~+$2.5M modeled. 3-year ROI: modeled 240%.
🔧	📈	🚀

Internal scenario — 13–15% of AI projects achieve these modeled returns.

Impact Dashboard: A Directional Cohort Checkpoint

💳

Monthly Copilot Cost

Per developer, calculated from AI-credit consumption. Tracks spend as agent-first cohort expands.

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Modeled Payroll Share

From a selectable salary input — a model input, not payroll data. Results are directional.

🔀

Pull Requests / Developer

Average PRs per developer per month. Throughput signal for early-phase vs agent-first cohorts.

⚠️ Dashboard results are potential and directional — not realized savings or causal proof. Access requires the Copilot usage metrics policy and View Copilot Metrics permission. Corrected 28-day cohort counts apply to the dashboard only, not the usage API or NDJSON exports.

Three Leadership Asks for Monday

Fund Instrumentation

Tests, security scanning, and quality gates must come before agents are deployed.

- 80%+ coverage on critical paths
- CI/CD quality gates
- Security scanning automated

Mandate Governance

IBM data confirms $2.2M in breach-cost savings for organizations that mandate governance first.

- Sanctioned model policy
- Access and audit controls
- Shadow AI detection in CI

Commit to Workflow Redesign

Agents in manual workflows deliver a fraction of the modeled return. Redesign is the multiplier.

- Redesign around agent capabilities
- Remove approval bottlenecks
- Measure iteration counts, not only speed

🎯 The question is whether to mandate measurement infrastructure — not whether to approve an AI pilot. The two decisions have different ROI profiles and different failure modes.

The Compounding Advantage Accrues to Organizations That Start Now

Platform maturity, team competency, and measurement infrastructure all compound over time

18–24 mo

head start in platform maturity and team competency for organizations that have already started

Internal scenario modeling — reflects infrastructure investment timeline and organizational learning curve

🏗 Platform Maturity Compounds

Infrastructure scales with headcount. Marginal cost per additional engineer approaches zero.

🧠 Team Competency Compounds

Agents improve with richer context. Documentation and test coverage increase agent output quality.

📊 Measurement Infrastructure Scales

The verification pipeline built for 50 engineers works for 500. The investment amortizes at scale.

💡 The mandate is measurement infrastructure. The ROI timeline begins when instrumentation begins — not when the first agent is deployed.

📊 The Value Recap

From Labor Arbitrage Theory to P&L Reality

Before

- Agent cost advantage identified but no mechanism to capture it
- AI pilots approved but P&L impact unmeasured
- Developers subjectively faster with no objective baseline
- Infrastructure investment deferred as optional cost

After

- Verification pipeline creates automated trust for agent output
- Issue lifecycle automation delivers $1M annual savings (scenario)
- Measurement surfaces where gains are real and repeatable
- Instrumentation investment pays back in 12–18 months (modeled)

20–65×
labor cost differential — agent compute vs senior engineer

$1M/yr
savings from issue lifecycle (50-person team, modeled scenario)

240%
modeled 3-year ROI with proper infrastructure investment

✅ What You Can Do Today

Today

- Read the Copilot impact dashboard docs on cohort access and permission requirements
- Calculate current issue lifecycle cost: issues/week × hours/issue × loaded hourly rate
- Identify one high-volume, approval-light workflow as the pilot candidate

This Week

- Map the five infrastructure barriers against the current engineering environment
- Request the Copilot usage metrics policy and View Copilot Metrics access
- Scope a 4–6 hour setup for issue triage automation in one repository

This Month

- Run the issue lifecycle pilot and measure time-to-triage before and after
- Build the business case using the internal modeled J-curve (Year 1 ~-$425K)
- Present three leadership asks: fund instrumentation, mandate governance, commit to workflow redesign

🔑 Key Takeaway

Measurement infrastructure is the prerequisite — instrumentation investment precedes agent deployment for reliable ROI.

References

📊 Research & Benchmarks

[McKinsey QuantumBlack: State of AI 2024 — 5% material improvement rate]
Foundational benchmark for AI pilot P&L success rates

[METR RCT 2025 — developers 19% slower on complex tasks with AI]
Perception vs measurement gap on complex real-world codebases

[DORA: ROI of AI-Assisted Development, 2025]
Organizational adoption vs output trust gap data

[Yetistiren et al.: AI code correctness — Copilot 46%, arXiv 2023]
Benchmark correctness rates for AI code generation tools

[GitHub + Accenture: 84% more successful builds, 2024]
Enterprise RCT on structured agentic development workflows

🔗 Related Talks

[The Agentic Labor Multiplier]

💰 Cost & Economics Data

[IBM Cost of a Data Breach, 2025 — $4.44M avg, $2.2M AI security savings]
Governance ROI data; 20% of 2025 breaches from shadow AI

[OpenAI API Pricing, 2025 — basis for $2–5/hr agent cost range]
Agent compute cost foundation for the labor arbitrage model

[BLS Occupational Employment: Software Developers, 2024]
Baseline for fully-loaded engineer labor cost ranges

[GitHub Changelog: Copilot impact dashboard ROI section, 2026]
Directional dashboard for cohort cost and PR volume comparison

Agentic Economics

The Agentic Economics: Making the Business Case

20–65×	5%	3.6 days	240%
labor cost differential — agent compute vs senior engineer (BLS 2024)	of AI pilots deliver material P&L improvement — the capture problem is infrastructure	payback on issue lifecycle automation — the highest-ROI entry point (modeled)	modeled 3-year ROI — contingent on instrumentation investment coming first

What is the biggest infrastructure gap preventing the organization from capturing the $2–5/hour opportunity today?